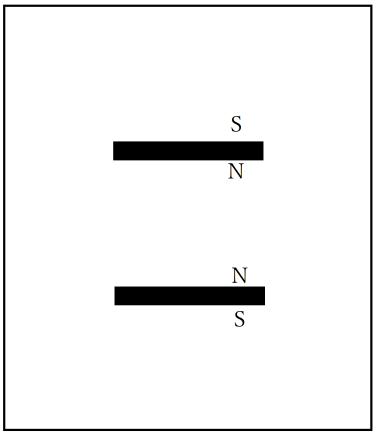
Mass Measurement (10 points)

Write down the numbers 0 to 9 in the following table:

0	1	2	3	4	5	6	7	8	9

Part A: Hooke's law and electromagnetic forces (2.4 points)

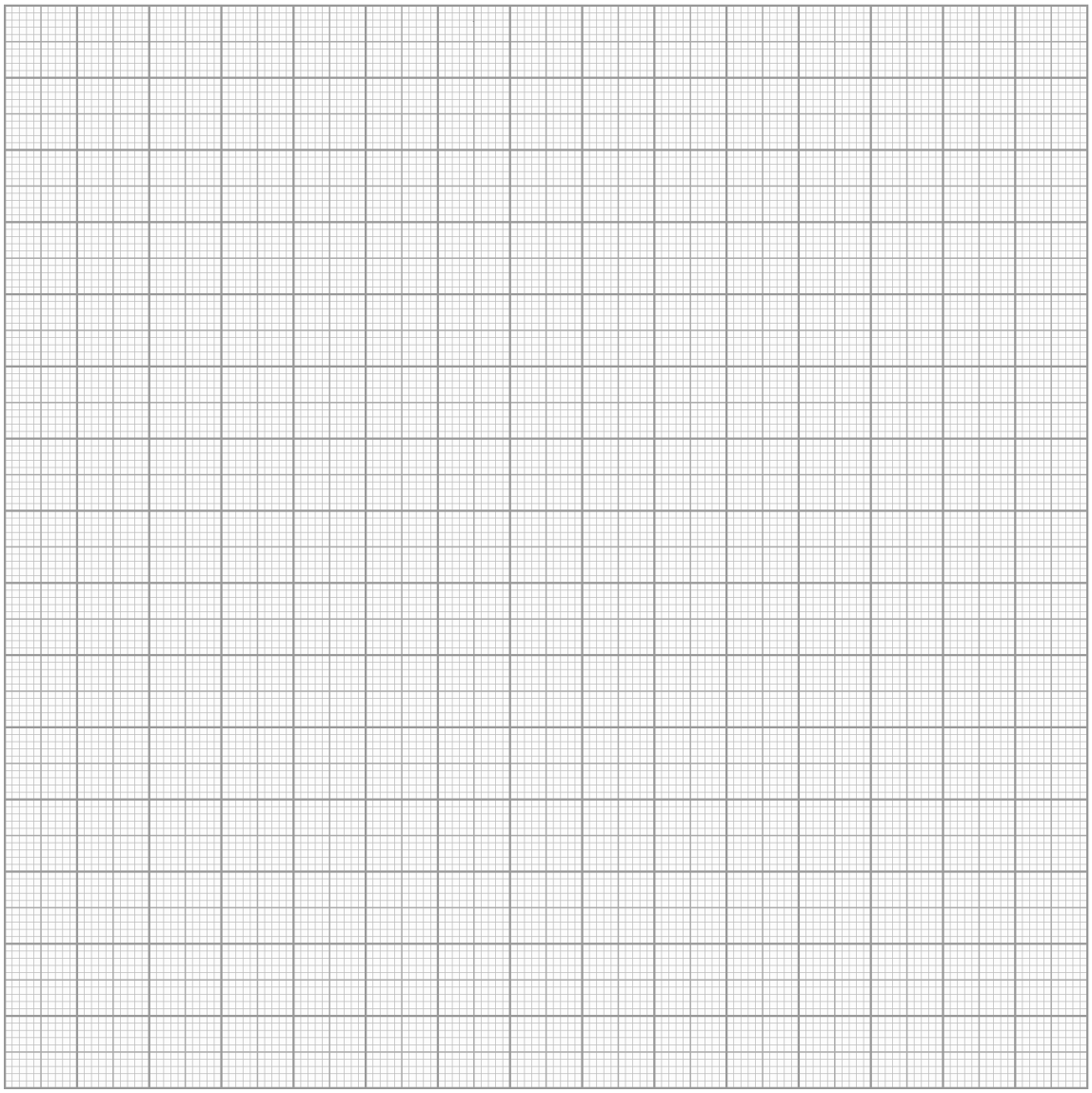
A.1 (0.4pt)



A.2 (0.6pt)

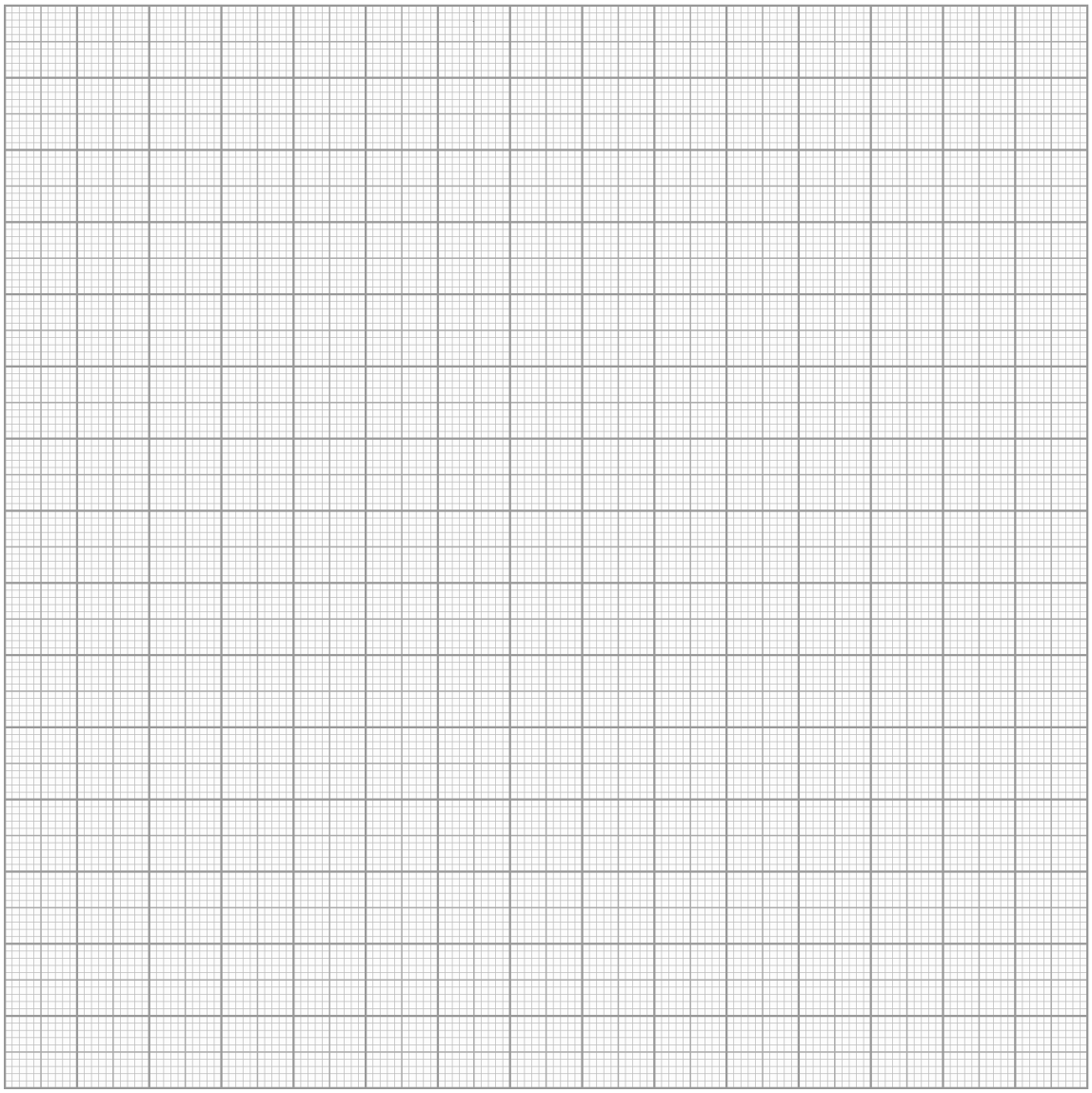
N	z /mm	I /A
0		0
1		
2		
3		
4		
5		

A.3 (0.7pt)



$a =$

A.4 (0.7pt)



$b =$

Part B: Induced electromotive force (3.0 points)

B.1 (0.2pt)

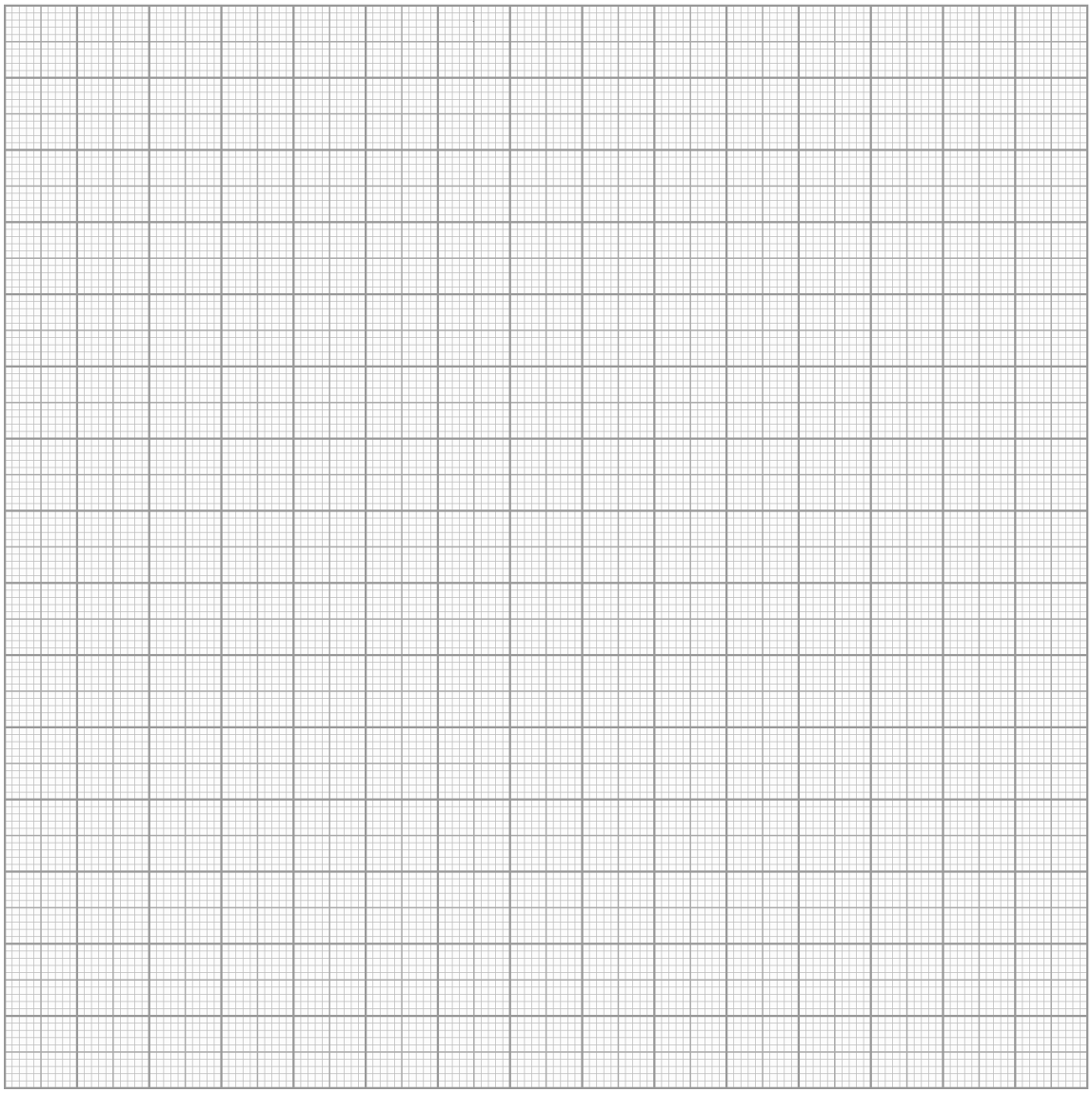
$V =$

B.2 (0.5pt)

$f_{\mathrm{B}} =$

A /mm	V' /V

B.3 (0.7pt)



$c =$

B.4 (0.4pt)

$BL =$

B.5 (1.2pt)

$m =$

$k =$

Part C: Mass dependence of resonant frequency (2.3 points)

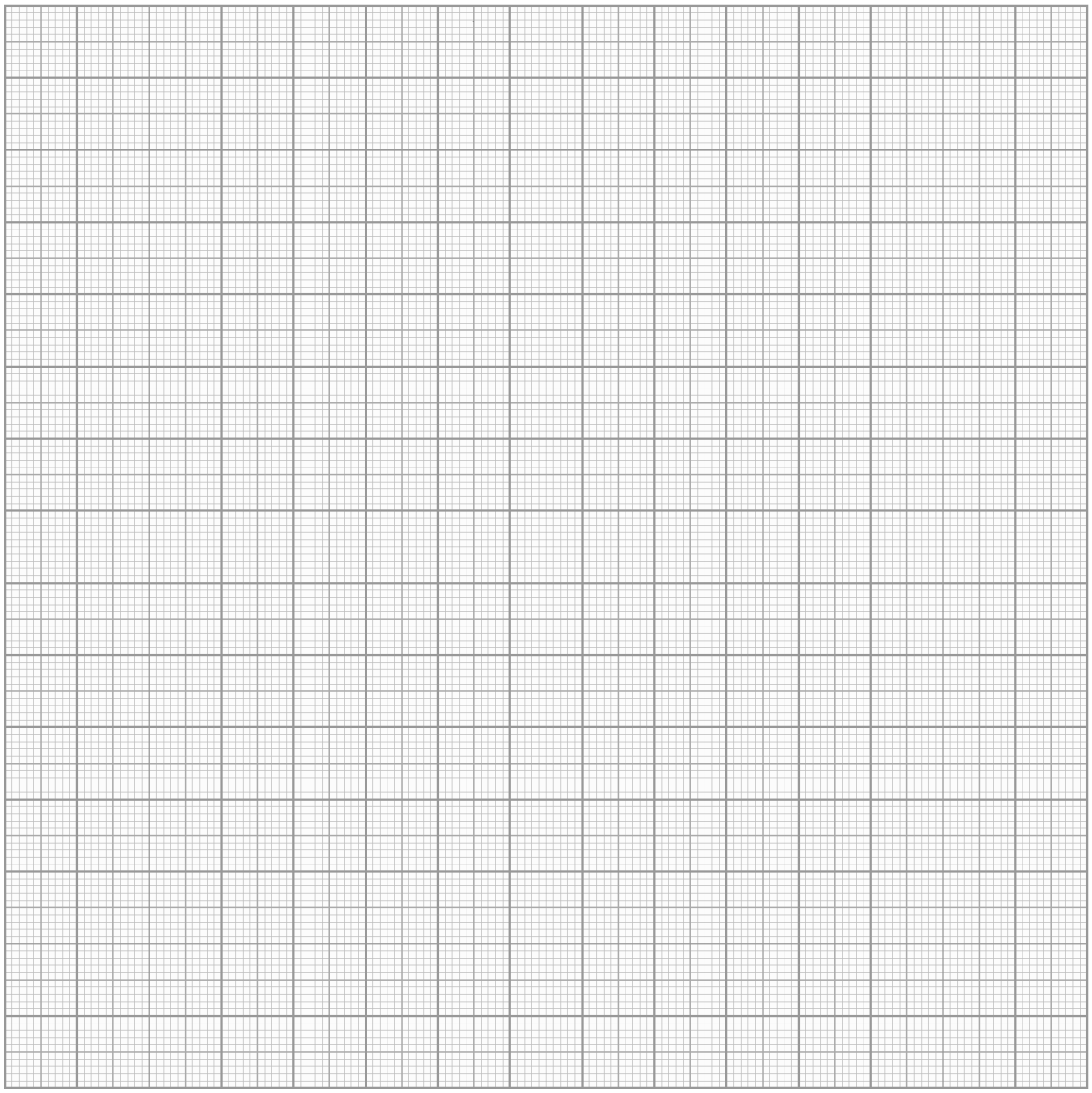
C.1 (0.2pt)

$f =$

C.2 (0.5pt)

N	f /Hz			
0				
1				
2				
3				
4				
5				

C.3 (1.0pt)



$\frac{M}{k'} =$

$\frac{m}{k'} =$

C.4 (0.6pt)

$\frac{M}{m} =$

$M =$

$k' =$

Part D: Resonance characteristics (2.3 points)

D.1 (0.4pt)

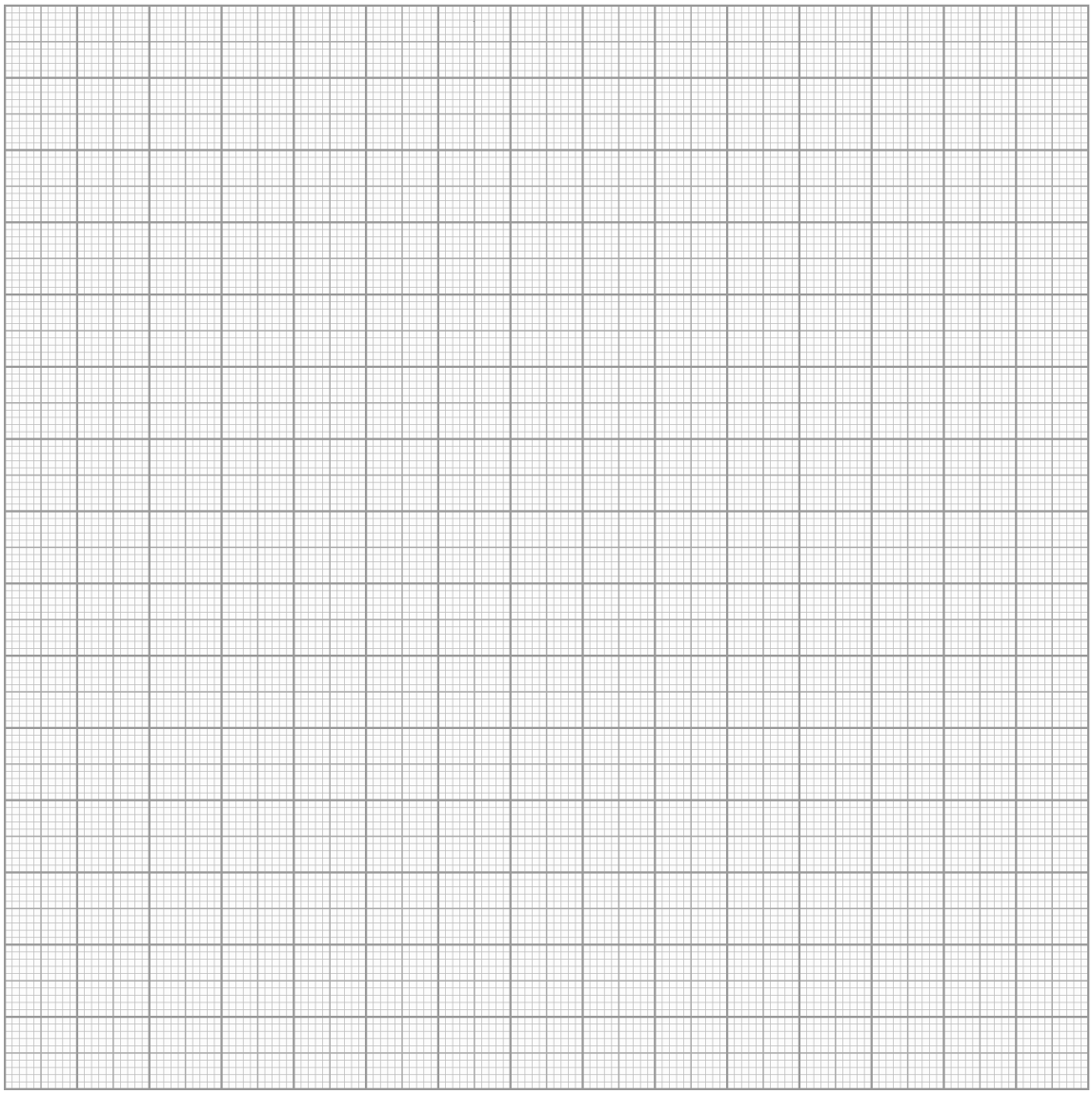
$V'_{\mathrm{AC}} =$

$F_{\mathrm{AC}} =$

D.2 (0.9pt)

f /Hz	A /mm		

D.2 (cont.)



D.3 (1.0pt)

$M =$